

Quiz 8

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (4 points) Find an equation for the tangent plane to the surface $z = xy$ at the point $(1, 1, 1)$.

Solution: $f(x, y, z) = xy - z$

$$\nabla f(x, y, z) = \langle y, x, -1 \rangle \Rightarrow \nabla f(1, 1, 1) = \langle 1, 1, -1 \rangle$$

Thus an equation for the tangent plane is

$$f_x(1, 1, 1)(x-1) + f_y(1, 1, 1)(y-1) + f_z(1, 1, 1)(z-1) = 0$$

$$(x-1) + (y-1) - (z-1) = 0$$

$$x + y - z = 1$$

Problem 2. (6 points) Find the local maximum and minimum values and saddle points of the function $f(x, y) = x^4 + y^4 - 4xy + 2$.

Solution: $f_x(x, y) = 4x^3 - 4y$, $f_y(x, y) = 4y^3 - 4x$

$$\begin{cases} 4x^3 - 4y = 0 \\ 4y^3 - 4x = 0 \end{cases} \Rightarrow \begin{cases} x^3 - y = 0 \\ y^3 - x = 0 \end{cases} \Rightarrow \begin{cases} y = x^3 \\ x = y^3 \end{cases} \Rightarrow x^9 - x = 0$$

$$\Rightarrow x(x^8 - 1) = 0 \Rightarrow x(x-1)(x+1)(x^2+1)(x^4+1) = 0$$

$$\Rightarrow x = 0, x = 1, x = -1$$

$$\begin{matrix} \downarrow & \downarrow & \downarrow \\ y = 0 & y = 1 & y = -1 \end{matrix}$$

Critical points: $(0, 0)$, $(1, 1)$, $(-1, -1)$

$$f_{xx} = 12x^2, f_{yy} = 12y^2, f_{xy} = -4$$

$$\text{The discriminant: } D(x, y) = (12x^2)(12y^2) - (-4)^2 = 144x^2y^2 - 16$$

i) $D(0, 0) = -16$, thus $(0, 0)$ is a saddle point.

ii) $D(1, 1) = 144 - 16 = 128 > 0$, $f_{xx}(1, 1) = 12 > 0$, thus

$f(1, 1) = 1 + 1 - 4 + 2 = 0$ is a local minimum

iii) $D(-1, -1) = 144 - 16 = 128 > 0$, $f_{xx}(-1, -1) = 12 > 0$, thus

$f(-1, -1) = 0$ is also a local minimum.